

Soluble Reactive Phosphorus

- An easy way to measure SRP using a spectrophotometer

Materials

PPE

- Lab coat
- Goggles
- Nitrile Gloves

Chemicals

- Concentrated Sulfuric Acid
- Deionized water
- Potassium Antimony Tartrate
- Ammonium molybdate $\circ 4\text{H}_2\text{O}$
- Ascorbic acid

Preparation

Note: Phosphate Reagent is stable for *4 hours*. All reagents can be kept at room temperature, except Ascorbic acid which should be stored at 4°C .

5N Sulfuric Acid

- Mix 70mL conc. and 500mL water

Potassium Antimony Tartrate

- Mix 0.2742g Potassium Antimony Tartrate with 100mL water

Ammonium molybdate $\circ 4\text{H}_2\text{O}$

- Mix 4g Ammonium molybdate $\circ 4\text{H}_2\text{O}$ with 100mL water

Ascorbic Acid

Stable one week

- Mix 1.76g Ascorbic acid with 100ml water

Combined Reagent

This is enough for 50 samples and standards

1. Mix 50mL 5N Sulfuric acid with 5mL Potassium Antimony Tartrate
2. Mix in 15mL Ammonium molybdate
3. Add 30mL Ascorbic Acid

Procedure

Standards

Standards should be prepared as in the table below. Depending on your sample concentrations, you may need a higher standard, or you may not need all of the standards below. Working standard concentration is below, you will have to dilute the 100 mg/L PO₄ stock to make this stock when you are ready to run your samples.

	H ₂ O Volume (mL)	Working Stock Volume (mL)	Target Concentration ($\mu\text{g L}^{-1}$)
1	10	0	0
2	10	0.10	3.10
3	10	0.20	6.10
4	10	0.40	11.90
5	10	1.00	28.20
6	10	2.00	51.70
7	10	4.00	88.60

Protocol

1. Pipette 10 ml of DI water into 14 tubes or vials. Add stock in amounts in standard table above.
2. Pipette 10 ml of filtered sample into separate test tubes.

3. Turn on spec.
4. Mix Combined reagent and Ascorbic Acid in correct proportions. Make enough for 2 ml per sample and standard, rounding up to nearest 100 ml to allow for mistakes. When Ascorbic acid is added the solution should turn a light yellow. Sometimes it looks a little yellowish-green
5. Add 2 ml of phosphate reagent to each tube and vortex to mix.
6. Let color develop for at least 10 minutes (20 would be better).
7. Read absorbance on a spectrophotometer at 880 nm.
8. When calculating standards, use both slope and intercept.